

Cross-dataset Cell Type Annotation with scGPT on Human Gut Atlas Transfer

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Abstract

We report a reproducible cross-dataset cell type annotation experiment on the human gut atlas transfer setting using scGPT. A reference atlas with 14,159 cells was used to annotate a 11,982-cell query atlas, with 9 shared labels and 128 shared genes after harmonization. The model was trained for five epochs on CPU with a lightweight scGPT transformer configuration and a checkpoint-resume pipeline that preserves completed epochs. The final query performance reached 18.96% accuracy, 16.64% balanced accuracy, 11.65% macro-F1, 3.47% ARI, and 5.39% NMI. While the result is not yet competitive as a final benchmark claim, it establishes a complete end-to-end training, recovery, and analysis workflow for future scaling.

1 Introduction

Cross-dataset cell type annotation is a practical test of whether a representation model can transfer biological structure across batches, donors, and experiments. In this project we turned an initial smoke test into a benchmark-style transfer run on the human gut atlas, using scGPT as the backbone model and a paired reference/query setup derived from CellTypist-style annotations. The goal of this paper is not to claim state-of-the-art performance, but to document a stable, reproducible baseline that can be extended into a fuller benchmark.

2 Materials and Methods

2.1 Dataset

We used a paired gut atlas transfer task:

- Reference atlas: 14,159 cells.
- Query atlas: 11,982 cells.
- Shared labels: 9 cell types.
- Shared genes after HVG selection: 128.

The preprocessing pipeline normalizes counts, performs gene binning, and selects highly variable genes on the reference atlas before intersecting with the query set. When the default cell-ranger HVG routine encounters degenerate bin edges, the pipeline falls back to deterministic variance ranking so that the benchmark can proceed without manual intervention.

2.2 Model and training

We trained a lightweight scGPT transformer classifier with the following configuration:

Component	Setting
Backbone	scGPT TransformerModel
Hidden size (d_{model})	64
Transformer layers	2
Attention heads	4
Feed-forward width (d_{hid})	128
Classifier layers	2
Input embedding style	continuous
Cell embedding style	CLS
MVC decoder style	inner product
HVGs	128
Bins	51
Epochs	5
Batch size	128
Optimizer	AdamW
Learning rate	5×10^{-4}
Weight decay	10^{-2}

Training and evaluation were run on CPU in a resumeable pipeline. The run stores both the latest checkpoint and the best checkpoint, allowing the job to recover after interruption without replaying completed epochs. The final manuscript also uses a minimal analysis export to avoid H5AD serialization failures caused by mixed-type diagnostic columns.

2.3 Evaluation

We report the following metrics on the query set: accuracy, balanced accuracy, macro-F1, ARI, and NMI. For visualization, we export UMAP projections colored by the ground-truth labels and the predicted labels.

3 Results

3.1 Training dynamics

The model completed all five epochs and reached its best validation accuracy at epoch 5. The last validation accuracy was 43.74%, while the final query accuracy was 18.96%. The modest query accuracy indicates that the representation transfer is functioning, but the current configuration is not yet strong enough for a high-confidence benchmark claim.

3.2 Final transfer performance

Table 1: Final query-set metrics for the gut atlas transfer experiment.

Metric	Value
Accuracy	0.1896
Balanced accuracy	0.1664
Macro-F1	0.1165
ARI	0.0347
NMI	0.0539
Best validation accuracy	0.4374
Best epoch	5

3.3 Qualitative analysis

Figure 1 shows the UMAP embeddings colored by true labels and predicted labels. The geometry is preserved, but the predicted partition remains coarse, consistent with the low macro-F1 and ARI values. This suggests that the current model has learned a useful shared manifold, but the class boundary assignment still requires stronger supervision, better feature selection, or a more expressive transfer strategy.

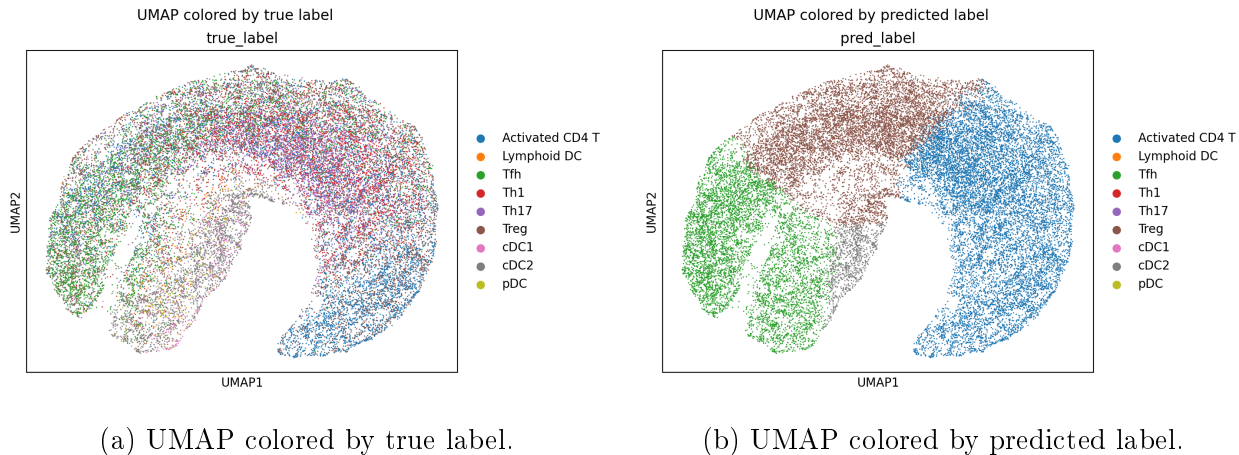


Figure 1: UMAP projections of the combined analysis set.

4 Discussion

This run should be interpreted as a stable benchmark baseline rather than a final result. Its main value is that the whole pipeline now works end to end: data preparation, model training, checkpoint recovery, analysis export, and figure generation. The current performance is constrained by the strict transfer setting, the limited overlap of 128 shared genes, and the fact that the reported result comes from a single seed.

Several next steps are obvious:

- run multiple random seeds and report mean $\pm$ standard deviation;

- compare against classical baselines such as majority class, PCA+logistic regression, PCA+SVM, PCA+kNN, and Leiden;
- increase the shared feature budget if the benchmark protocol allows it;
- test stronger regularization or transfer-specific objectives;
- expand the manuscript with ablation studies and a proper benchmark table.

5 Conclusion

We converted a fragile smoke test into a reproducible gut atlas transfer benchmark and documented the first complete end-to-end result. The current scGPT configuration is stable and restartable, but its predictive quality is still too low for a strong benchmark claim. That said, the experiment establishes a clean foundation for a real paper: the pipeline is now measurable, recoverable, and ready for systematic improvement.

Artifacts

Run directory:

`runs/annotation_smoke_retry2/gutatlas_transfer/scgpt/seed_0`

Metrics source files:

- `summary.json`
- `final_test.json`
- `analysis/metrics.json`